



Title | 2023 Social Trend

My part | Disability & Universal Design: Reinterpreted Through Universality and Market Logic

Meaning | This book, written in 2022, was an effort to establish my perspective on disability. While serving in the military, I felt disconnected from the evolving societal trends and was uncertain about how to apply service design to the context of disabilities. At that time, a friend introduced me to this publishing project, which focused on social trends and involved contributions from a diverse group of graduates.

Through this project, I had the opportunity to refine my thoughts on disabilities by engaging with scholars from various fields. Although my initial thoughts were somewhat immature and have since evolved into more developed theories, this work still represents my very first true attempt—a declaration of my intentions. The sincerity of my feelings at that time remains here. Therefore, I am pleased to share its English translation here. For those who may be short on time, I have also provided a brief summary. If you do find the time to read it, I would be truly grateful, as I hope my sincerity and passion for solving issues related to disabilities come across to you fully.

Abstract |

This book offers an expanded perspective on disability, emphasizing that disability is not limited to traditional forms such as visual or auditory impairments. It also includes situations like the inability to communicate in a foreign language or being unable to move due to an injury. These seemingly minor issues, often overlooked in everyday life, can also be considered disabilities for someone. The book argues that since such incidents can happen to anyone, disability should be addressed in a way that includes both non-disabled and disabled individuals.

The book explains the seven principles of Universal Design with specific examples for each. It also compares Barrier-Free Design and Universal Design, advocating for a shift from seeing disability through the lens of charity and welfare to addressing it as an issue to be 'embraced' through market-driven solutions from the outset. People with disabilities, like anyone else, desire to use beautiful things, highlight their strengths, and conceal their weaknesses. Therefore, no matter how practical a product may be, if it does not evoke a sense of desire, it cannot succeed in the marketplace.

This book suggests that instead of focusing solely on considerations for the vulnerable, designs should be thoroughly market-oriented—offering 'desirable' products that people want to buy. By incorporating Universal Design, the book envisions a world where disabilities are not differentiated from other customer needs, thereby expanding the market to include everyone naturally.

An example presented in the book is the comparison between hearing aids and wireless earbuds. Traditionally, hearing aids have made otherwise invisible hearing impairments visible. However, with the popularization of wireless earbuds, the distinction between the two has blurred. For instance, Company A introduced features like 'Live Listening' and 'Conversation

Boost' into their wireless earbuds to offer functionalities similar to hearing aids. According to research, while these earbuds may not match premium hearing aids in performance, they provide similar effectiveness to general hearing aids at a significantly lower cost. Premium hearing aids cost around \$10,000, whereas Company A's earbuds are priced at just \$249. Thus, if wireless earbuds fully absorb the functionalities of hearing aids, they will no longer be tools exclusively for people with disabilities. The potential of Universal Design, explained through market logic, is to create non-stigmatizing, desirable products that blend into everyday life, ultimately broadening the market.

The book goes on to discuss how this potential is not limited to people with disabilities. The elderly, individuals with temporary mobility issues, people who need to carry medical devices, and those who are inherently frail all represent customers with unique characteristics and an expanded market. Products designed with these individuals in mind can diversify the market, offering more choices and better quality of life. Thus, if dramatic treatments or miraculous surgeries cannot instantly eliminate disabilities, the next best approach is to gradually make the world a better place.

The book also argues that considering minorities that were previously overlooked in the pursuit of broader markets not only makes the world better but also presents new business opportunities. It cites examples of prosumer policies adopted by companies like Company S and Company L to seize these opportunities. Additionally, it introduces architectural examples like Dongdaemun Design Plaza (DDP) and Korean government policies such as BF Certification as ways to apply Universal Design more broadly.

The book concludes with the following message:

We all experience different perspectives and positions throughout our lives. In this journey, everyone is strong in some areas and weak in others. Not only individuals with disabilities but also those without disabilities have their vulnerabilities depending on the situation. If we can create an environment that compensates for these weaknesses from various perspectives, we can naturally become more empathetic and considerate of others, thereby enhancing our quality of life.

People can acquire disabilities at any point—from birth, as they age, due to accidents, illness, or complications. Disabilities can occur suddenly, unexpectedly, and for unforeseen reasons. Thus, having a disability is not a fault; it is simply another form of vulnerability.

Therefore, the application of Universal Design should not be understood from the perspective of charity but from a universal viewpoint that acknowledges it could one day be relevant to any of us. If we fill the world with diverse products and services, not merely focusing on policies or welfare but making Universal Design a fundamental part of the marketplace, then disability will not be highlighted but will instead be accepted as a natural part of society. In doing so, disabilities can blend more easily into the fabric of everyday life. I sincerely hope that one day this vision will become our reality.

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Disability & Universal Design
:Reinterpreted through Universality and Market Logic

Universal Design

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Disability can be defined in numerous ways, but to put it simply, it refers to "being restricted from fully performing certain functions." According to this definition, not only do conditions like visual, auditory, and physical impairments count as "disabilities" in the conventional sense, but situations like being unable to communicate in a foreign language while abroad, or being unable to climb to high places due to an injured leg, can also be considered disabilities. When we broaden our perspective in this way, even small things that we don't usually notice in our daily lives can become obstacles for someone else. And such obstacles have always been perceived as something to be overcome. But is it really necessary to always overcome disabilities? Could there be a wiser way to deal with them?

Let me provide an example. During the 7th year of King Sejong's reign in the past, there was an incident during a royal ancestral rite held at Jongmyo Shrine. Heo-Jo, the

Minister of Personnel, lost his footing and tumbled down the stairs. When Heo-Jo asked for forgiveness, King Sejong responded:

"Minister Heo-Jo, are you hurt? It seems the stairs are too narrow; make sure they are widened."

Heo-Jo's accident occurred because he had difficulty moving due to a congenital spinal disability. However, regardless of Heo-Jo's disability, King Sejong's words hold significant meaning. Rather than blaming Heo-Jo's disability, King Sejong pointed out that anyone could have such an accident, and he suggested a solution that would prevent accidents for both people with and without disabilities.

Recognizing that no matter how trivial something might seem, it can become an obstacle for someone, and finding solutions that consider everyone—this is how we ensure no one is discriminated against and that everything is accessible to all. This is what Ronald Mace, an American architect and industrial designer, articulated and promoted in 1997 as "Universal Design."

Universal Design and the 7 Principles

Universal Design is a concept initially proposed by British architect and human rights advocate Selwyn Goldsmith in his book 'Designing for the Disabled', and later developed by Professor Ronald Mace at the University of North Carolina. It refers to "Design for All"—a design that ensures no discrimination occurs in the use of products, services, or spaces, regardless of disability, age, language, or culture. Let's consider an example.

There are countless products in the world, but most are designed primarily for the majority—right-handed people. A prime example of this would be scissors or guns. The discomfort that right-handed people experience when using scissors with their left hand is exactly the discomfort that left-handed people feel every time they use scissors designed for right-handed individuals. To address this problem, "ambidextrous scissors" were introduced, allowing both right- and left-handed people to use them without feeling discriminated against. This is also an example of Universal Design.

Universal Design is realized through its foundational seven principles.

1. Equitable Use
2. Flexibility in Use
3. Simple and Intuitive Use
4. Perceptible Information
5. Tolerance for Error
6. Low Physical Effort
7. Size and Space for Approach and Use

The first principle, 'Equitable Use', emphasizes that everyone should be able to use a product without feeling discriminated against, anxious, or inferior. It aims to provide the same or equal means of use for all users. For example, automatic doors fulfill this principle by opening when approached by anyone, regardless of their characteristics, ensuring 'equitable use'.

The second principle, 'Flexibility in Use', requires that products be usable accurately and freely under various environmental conditions. One example is a pen designed with different physical conditions in mind. This pen has a U-shaped bend in the middle with a hole that allows the user to hold it without needing all five fingers. Thanks to this loop, users can hold and use the pen with fewer

fingers, or even with their feet or mouth, thus offering multiple options and satisfying ‘flexibility in use’.

The third principle, ‘Simple and Intuitive Use’, involves making products easy to understand with clear usage feedback, eliminating unnecessary complexity. A common example is the emergency alarm bell found in buildings. Its red design, along with large letters saying ‘Emergency Bell’, allows users to intuitively understand that it is meant to be pressed in emergencies. Additionally, pressing the bell provides immediate feedback through an alarm, fulfilling the requirement for ‘simple and intuitive use’.

The fourth principle, ‘Perceptible Information’, is about conveying essential information clearly through various means such as visuals, sound, and touch, making it as easy to perceive as possible. An example is tactile blocks on the floor for visually impaired individuals, often designed in yellow and raised so that they are easy for both visually impaired and non-impaired people to recognize. Similarly, tactile maps that incorporate visual and tactile elements provide information to a wide range of users.

The fifth principle, ‘Tolerance for Error’, involves arranging elements in a way that minimizes hazards and mistakes, and allows easy recovery from errors. One example is the QWERTY keyboard, designed to prevent

typewriter jams when typing too quickly by placing frequently used keys farther apart. The keyboard's undo function is another example of 'tolerance for error'.

The sixth principle, 'Low Physical Effort', ensures that products can be used without excessive force or repetitive motion, allowing for a natural and comfortable posture. Many of us have experienced the difficulty of unplugging electrical plugs. To solve this problem, some plugs have a hole in the middle for inserting fingers, allowing for easy removal with less effort, demonstrating the principle of 'low physical effort'.



Fig.1 Example of Applying the Principle of 'Low Physical Effort'

The final principle, 'Size and Space for Approach and Use', refers to ensuring that items are easy to access or store, and that all objects are placed so they are comfortably reachable for both seated and standing users. Additionally, adequate space should be provided for the use of assistive devices or for receiving assistance from others. A prime example of this is accessible restrooms for people with disabilities.

Universal Design and Barrier Free

As we explore these examples, some may think of a similar concept called "Barrier-Free." Like Universal Design, Barrier-Free is an approach that aims to overcome the inconveniences caused by disabilities. The term Barrier-Free originates from a report titled 'Barrier-Free Design', which was submitted at the UN's Expert Group Meeting on Disability and the Living Environment in 1974. This design approach focuses primarily on architecture, aiming to remove physical barriers that prevent individuals with disabilities from accessing spaces. Examples of Barrier-Free applications include wheelchair lifts installed on subway staircases and ramps in place of stairs. Initially, Barrier-Free focused on removing physical obstacles in buildings, but the concept gradually expanded into other

areas, such as Barrier-Free Films that incorporate captions and audio descriptions for individuals with visual or hearing impairments.

Barrier-Free is a concept that was advocated before Universal Design and serves as its root. However, there are significant differences between Barrier-Free and Universal Design.

Barrier-Free design introduces specific solutions to remove 'barriers' that individuals with disabilities face in everyday life, aiming to address their 'special needs' and overcome the limitations caused by disability. The significance of this approach lies in its focus on overcoming disability. However, this approach also has its limitations—it tends to highlight the disability itself. For instance, the wheelchair lifts found in subway stations are a representative example of Barrier-Free solutions intended to overcome the limitations faced by wheelchair users on stairs. However, these lifts are often met with negative feedback due to their open structure, alarms, and flashing lights, which attract attention and end up emphasizing the individual's disability.

In contrast, Universal Design features elevators in subway stations that enhance convenience for everyone, regardless of age or disability, naturally integrating people

with disabilities as part of the larger community.

Both Barrier-Free and Universal Design aim to address disability and propose measures for it, but why do they differ in their approach? This difference stems from their underlying perspectives—whether they aim to ‘overcome’ a problem by resolving it, or ‘embrace’ it from the start as if it doesn't exist. In other words, while Barrier-Free may unintentionally draw attention to disability during the process of solving a problem, Universal Design seeks to make disability irrelevant by striving for inclusive use from the outset.

This implies that Barrier-Free designs might fail to meet the ‘desire to use’ that consumers have, while Universal Design products are more likely to satisfy that desire.

Universal Design Explained Through Market Logic

From a market perspective, individuals with disabilities are customers before they are people with disabilities. Like any other customer, they have desires that should be understood and fulfilled. They, too, want to use beautiful things, highlight their strengths, and hide their weaknesses.

No matter how practical a product may be, if it doesn't make you want to use it, we can't ignore the basics of a market where purchasing power matters.

We need to reconsider whether we view individuals with disabilities merely as subjects of care and welfare because of their disabilities, because they are seen as "vulnerable." Before considering care and welfare, we need to go back to the basics of treating each person as a human being and a customer. To truly expand the choices for solving problems, we must go beyond the perspective of care for the vulnerable and focus on selling thoroughly market-driven, "desirable designs."

Moving towards Universal Design means eliminating the distinction between disabled and non-disabled people, including everyone—every customer—under the same category. In other words, individuals with disabilities become natural customers without being conscious of their disabilities, which, in turn, expands the market significantly.

Let's take a specific example. Hearing aids make the otherwise invisible auditory disability visible by wearing them. For this reason, hearing aids have historically been something people were reluctant to use, despite their necessity. However, with the popularization of wireless

earbuds, hearing aids are no longer products that visualize disabilities. People walking down the street with wireless earbuds and those wearing hearing aids are no longer distinguishable from each other. For example, the promotional materials for hearing aid manufacturers now often include phrases like “looks just like the latest Bluetooth earphones”.

A leading brand of wireless earbuds, Brand A, includes features like Live Listening, which amplifies surrounding sounds through a smartphone microphone and sends them to the earbuds, and Conversation Boost, which uses the earbud microphone to amplify the voice of the person in front of, enhancing conversation volume. These features are intended for people with hearing impairments and have a similar effect to that of hearing aids.

According to research published in the Journal of Fusion Science ‘T’, a joint study by Taiwan's Hospital B, K University of Health Sciences, and K Medical University experimented with the possibility of replacing hearing aids with Brand A's wireless earbuds for 21 patients with mild to moderate hearing loss. The results showed that while Brand A's product performed worse than premium hearing aids in terms of noise in a quiet environment, its performance was comparable to standard hearing aids

without a prescription. Furthermore, it even outperformed premium hearing aids in noisy environments.

The results indicated that the price of premium hearing aids is \$10,000 (approximately KRW 13,350,000), and the price of standard hearing aids is \$1,500 (approximately KRW 2,000,000). In contrast, the price of Brand A's wireless earbuds is \$249 (approximately KRW 330,000), while the entry-level version costs only \$129 (approximately KRW 170,000).

As the researchers pointed out, the study is still in its early stages, and the intended purpose of wireless earbuds and hearing aids is strictly different. Therefore, it cannot be generalized that Brand A's product is necessarily as effective as hearing aids. However, if cheaper wireless earbuds are eventually released with performance comparable to hearing aids, accessibility to hearing aids could significantly improve. At the same time, from the perspective of the wireless earbud market, there will be new potential for a larger market by encompassing both disabled and non-disabled individuals.

What would happen if wireless earbuds perfectly absorbed the function of hearing aids? Assistive tools for auditory impairment would no longer be exclusively for those with hearing loss. Designing without discrimination,

creating products that don't cause reluctance to use, and allowing these tools to seamlessly blend into the new norm—that is how a larger market is created. This is the potential of Universal Design as explained through market logic.

The Expandability of Universal Design

This potential is not limited to disabilities alone. For example, in the case of the elderly, just as with people with disabilities, the elderly are also “customers” with the same purchasing power—it's merely that they are older. They are simply individuals who, with age, face a few more inconveniences and new challenges to solve.

Elderly people, patients who are temporarily immobile, those who need to carry medical devices at all times, or those who are frail from birth—all these are individuals with distinct characteristics, and they collectively represent an expanded market of possibilities. Products designed with these people in mind diversify the market and provide the potential for a broader range of choices and a better quality of life.

Consider the example of an inhaler redesigned to resemble an accessory, aiming to change perceptions so

that individuals who need to use it to prevent asthma exacerbation do not feel self-conscious or reluctant to use it in front of others. Or an electric shaver designed so that if shaving becomes difficult due to age, illness, or an accident, someone else can easily help. A wristwatch that allows the wearer to perceive the time through touch was inspired by the limitation of conventional voice-activated clocks in environments that need quiet, such as crowded places, lecture halls, or libraries. An induction cooktop redesigned to distinguish different cooking areas, heat settings, and timers through tactile elements, addressing the fact that conventional induction cooktops are just flat, hot, rectangular surfaces for people who can't see. A throwable fire extinguisher that is easy to use, created because traditional powder fire extinguishers are difficult for many people, especially children and the elderly, to handle properly.

If dramatic treatments or miraculous surgeries cannot instantly eliminate disabilities, gradually creating a better world is our best course of action. These products, developed by carefully considering even the smallest aspects and the perspectives of various users, are making the world a better place. Meanwhile, minority groups that were previously overlooked due to the focus on more

general markets are becoming new business opportunities. In practice, companies are beginning to implement a ‘prosumer’ policy, involving people with disabilities, elderly individuals, children, and others from diverse backgrounds from the very beginning of the product planning and development stages, under the belief that ‘if the most disadvantaged person can use it comfortably, then it will be easy for everyone to use’.

For instance, since 2016, as part of its prosumer policy, Company S has reflected the opinions of ‘S Supporters’, which includes individuals with disabilities, during the development of its smartphones. Through this process, they learned that sudden voice assistant guidance can be embarrassing for visually impaired users in public spaces, leading them to make improvements to eliminate sound when not needed. They also added a feature that detects doorbell rings or a baby's crying and notifies users with vibrations and visual alerts on the phone, considering the needs of hearing-impaired users.

In May 2021, Company L established an ‘Accessibility Advisory Committee’ consisting of 13 members, including 7 disabled experts chosen by the Korean Disabled Consumers Association and 6 evaluators with visual, auditory, and physical disabilities. This committee has been producing

and managing evaluation indicators for the accessibility and convenience of Company L's home appliances. The Seoul Tourism Foundation has also set up a 'Barrier-Free Kiosk' at the Seoul Tourism Plaza for people with limited mobility as part of its efforts to gather feedback and make improvements. Six individuals with disabilities participated in the user testing, including two representatives each with visual, hearing, and physical impairments. Overseas, individuals with disabilities are often employed as professional consultants or beta testers, working as advisory members for companies to provide insights on their products and services.

Measures for the Universal Application of Universal Design

Although Universal Design has already become more prominent today compared to the past, reflecting diverse perspectives, we must not become complacent but instead continue to make progress. For this, we need a stage where Universal Design can be more widely known and accepted. In South Korea, Dongdaemun Design Plaza (DDP) is an example of such a platform.

DDP, with its minimal use of stairs and open spaces without partitions, allows free movement for both

wheelchairs and strollers. You can comfortably sit anywhere, even without designated chairs, and the handrails in the corridors are designed to connect to the ground without interruptions. Thanks to this thoughtful design, DDP has been praised as a building that embodies Universal Design. Additionally, on the third floor's Design Lab, there is a 'Universal Design Lifestyle Platform (UDP)' where visitors can learn about and experience Universal Design. Companies like Company H, which makes clothing tailored to users' physical characteristics, Company I, a furniture store, and Company D, a home appliance brand, exhibit products incorporating Universal Design that people can try out.

To ensure that environments incorporating Universal Design are not limited to places like DDP, the government is attempting to create more favorable conditions through various forms of legislation. A representative example of this is the Barrier-Free(BF) Certification System. Operated jointly by the Ministry of Health and Welfare and the Ministry of Land, Infrastructure, and Transport, the BF Certification System aims to create a safe and convenient living environment that allows everyone—including individuals with disabilities, the elderly, and pregnant women—to easily access, use, and navigate local and

individual facilities. Consequently, all ‘public facilities and public-use facilities’ newly constructed by national or local governments must receive the ‘Barrier-Free Certification’. An evaluation committee made up of experts in related facilities and the field of welfare for people with disabilities works hard to reflect the real lives of individuals with disabilities, and the certification is valid for five years.

Starting from buildings like DDP, to cities like Seoul or Daejeon that have embraced this concept, if more environments adopt these standards, Universal Design will naturally become part of people's everyday lives. In such a scenario, a culture that considers, empathizes with, and reflects everyone's needs will be cultivated. Companies, in turn, will offer products and services designed with Universal Design to attract a wider variety of customers. This will eventually lead to the universalization of Universal Design.

Conclusion

As we go through life, we all experience different perspectives and positions. Throughout the course of life, everyone is strong in some areas and weak in others. Not

only individuals with disabilities but also those without disabilities each face their own obstacles depending on their circumstances. Therefore, if we can create environments that accommodate and compensate for weaknesses from diverse perspectives, we will naturally learn to consider others' viewpoints more frequently, leading to a better quality of life for all.

Anyone can acquire a disability. It could happen from birth, as we age, due to an accident, an illness, or complications. It can happen at any time without warning, even as suddenly as when you wake up the next day for an unexpected reason. Having a disability is not a fault; it is simply another form of vulnerability.

Thus, the application of Universal Design should not be understood from the perspective of 'charity' but rather from a 'universal' perspective, one that acknowledges that the need may someday be relevant to us. If we fill the world with diverse products and services, not just policies or welfare, and if Universal Design becomes a natural and essential part of the marketplace, only then can disability be accepted as something normal rather than highlighted. Only then can it blend into society more seamlessly. I sincerely hope that one day this becomes our reality.